

Diversity in Ensemble Feature Selection

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Abstract

Ensembles of learnt models constitute one of the main current directions in machine learning and data mining. Ensembles allow us to achieve higher accuracy, which is often not achievable with single models. It was shown theoretically and experimentally that in order for an ensemble to be effective, it should consist of high-accuracy base classifiers that should have high diversity in their predictions. One technique, which proved to be effective for constructing an ensemble of accurate and diverse base classifiers, is to use different feature subsets, or so-called ensemble feature selection. Many ensemble feature selection strategies incorporate diversity as a component of the fitness function in the search for the best collection of feature subsets. There are known a number of ways to quantify diversity in ensembles of classifiers, and little research has been done about their appropriateness to ensemble feature selection. In this paper, we compare seven measures of diversity with regard to their possible use in ensemble feature selection. We conduct experiments on 21 data sets from the UCI machine learning repository, comparing the ensemble accuracy and other characteristics for the ensembles built with ensemble feature selection based on the considered measures of diversity. We consider five search strategies for ensemble feature selection: simple random subsampling, genetic search, hill-climbing, ensemble forward and backward sequential selection. In the experiments, we show that, in some cases, the ensemble feature selection process can be sensitive to the choice of the diversity measure, and that the question of the superiority of a particular measure depends on the context of the use of diversity and on the data being processed.

Keywords: Ensemble of classifiers; Ensemble diversity; Feature selection; Search strategy; Dynamic integration of classifiers

1 Introduction

Current electronic data repositories are growing quickly and contain a huge amount of data from commercial, scientific, and other domain areas. This data includes also currently unknown and potentially interesting patterns and relations, which can be uncovered using knowledge discovery and data mining methods [13].

A popular method for creating an accurate classifier from a set of training data is to train several different classifiers, and then to combine their predictions. The integration of multiple classifiers, to

improve classification results, is currently an active research area in the machine learning and neural networks communities [2, 4 – 11, 14 – 17, 20 – 31, 34 – 43]. Dietterich [10] has presented the integration of multiple classifiers as one of the four most important directions in the machine learning research. It was shown in many domains that an ensemble is often more accurate than any of the single classifiers in the ensemble.

Both theoretical and empirical research have demonstrated that a good ensemble is one where the base classifiers in the ensemble are both accurate and tend to err in different parts of the instance space (that is have high diversity in their predictions). Another important issue in creating an effective ensemble is the choice of the function for combining the predictions of the base classifiers. It was shown that increasing coverage of an ensemble through diversity is not enough to ensure increased prediction accuracy – if the integration method does not utilize coverage, then no benefit arises from integrating multiple models [4].

One effective approach for generating an ensemble of accurate and diverse base classifiers is to use different feature subsets, or so-called *ensemble feature selection* [27]. By varying the feature subsets used to generate the base classifiers, it is possible to promote diversity and produce base classifiers that tend to err in different subareas of the instance space. While traditional feature selection algorithms have the goal of finding the best feature subset that is relevant to both the learning task and the selected inductive learning algorithm, the task of ensemble feature selection has the additional goal of finding a set of feature subsets that will promote disagreement among the base classifiers [27].

Ho [17] has shown that simple random selection of feature subsets may be an effective technique for ensemble feature selection because the lack of accuracy in the ensemble members is compensated for by their diversity. Random subsampling is used as a base in a number of ensemble feature selection strategies, e.g. GEFS [27] and HC [8].

Feature selection algorithms, including ensemble feature selection, are typically composed of the following components [1, 27]: (1) *search strategy*, that searches the space of feature subsets; and (2) *fitness function*, that inputs a feature subset and outputs a numeric evaluation. The search strategy's goal is to find a feature subset maximizing this function.

As it is acknowledged that an effective ensemble should consist of high-accuracy classifiers that disagree on their predictions, many fitness functions in ensemble feature selection include explicitly or implicitly both accuracy and diversity. One measure of fitness, which was proposed by Opitz [27], defines the fitness $Fitness_i$ of a classifier i corresponding to a feature subset i to be proportional to the classification accuracy acc_i and the diversity div_i of the classifier (in fact, it is the contribution of the

classifier i to the total ensemble diversity, which can be measured as the average pairwise diversity for all the pairs of classifiers including i):

$$Fitness_i = acc_i + \alpha \cdot div_i, \quad (1)$$

where α is the coefficient of the degree of the influence of diversity. This fitness function was also used in experiments in [39], and we use it in our experiments in this paper.

A common measure of classification accuracy is the percentage of correct classifications on the test data set, or average within-class accuracy, the average percentage of correct classifications within each class, if the class distributions are significantly uneven. The simple “percentage of correct classifications” measure is successfully used in the vast majority of cases, as it reflects well the classification performance. For diversity, the situation is not that straightforward – there are known a number of ways to measure diversity in ensembles of classifiers, and not much research has been done about the appropriateness and superiority of one measure over another.

In this paper, we consider different measures of ensemble diversity, which could be used as a component of the fitness function (1), and which could also be used to measure total ensemble diversity, as a general characteristic of ensemble goodness. The goal of this paper is to compare the considered measures of diversity in the context of ensemble feature selection for these two particular tasks. To our knowledge, in all the existent research papers, comparing different measures of ensemble diversity, as [22, 34], the only common way of the comparison is to analyze the correlation of those measures of diversity with different other ensemble characteristic, as the ensemble accuracy, the difference between the ensemble accuracy and the average base classifier accuracy, the difference between the ensemble accuracy and the maximal base classifier accuracy, etc. In contrast to that common way, in this paper we compare the measures of diversity using the wrapper approach, applying them as a component of the fitness function guiding the search in ensemble feature selection, and comparing the resulting accuracies and other ensemble characteristics.

The paper is organized as follows. In Section 2 we consider the general task of constructing an effective ensemble, review different techniques for generating the base classifiers, and especially ensemble feature selection, and present four strategies for ensemble feature selection. In Section 3 we consider the question of integration of an ensemble of classifiers and review different integration methods. In Section 4 we present seven different measures for diversity in classification ensembles. In Section 5 we present our experiments with these measures and conclude in Section 6 with a summary and further research topics.

2 An ensemble and its diversity

In this section, we consider: the question of what constitutes a good ensemble, effective techniques for generating the base classifiers in ensembles, the place of ensemble feature selection among them, and search strategies for ensemble feature selection, which we shall use in our experiments.

2.1 Ensemble goodness criteria and their interrelation

An ensemble is generally more accurate than any of the base classifiers in the ensemble. Both theoretical and empirical research have shown that an effective ensemble should consist of base classifiers that not only have high classification accuracy, but that also make their errors in different parts of the input space [4, 9, 15, 21, 41]. Brodley and Lane [4] show that the main objective when generating the base classifiers is to maximize the *coverage* on the data, which is the percentage of the instances that at least one base classifier can classify correctly. Achieving coverage greater than the accuracy of the best base classifier requires *diversity* among the base classifiers. Several researchers have presented theoretical evidence supporting this claim [15, 21, 41].

For example, Hansen and Salamon [15] proved that, if the average classification error rate for an instance is less than 50% and the base classifiers in the ensemble are independent in the production of their errors, then the expected error for that instance can be reduced to zero as the number of base classifiers included into the ensemble goes to infinity, when majority voting is used for integration. Such assumptions rarely hold in practice (for example, an outlier may easily have predicted classification error rate that is higher than 50%), and then the classification error rate over all the instances cannot necessarily be reduced to zero. But if we assume a significant percentage of the instances are predicted with less than 50% average error, gains in generalization will be achieved. A key assumption in this analysis is that the base classifiers should be independent in their production of errors.

Krogh and Vedelsby [21] have shown that the classification error for an ensemble of neural network base predictors is related to the generalization error of the base networks and to how much disagreement there is between them. They have proved that $E = \bar{E} - \bar{A}$, where E is the *ensemble generalization error*, $\bar{E}$ is the average of the generalization errors of the base networks weighted by corresponding beliefs, and $\bar{A}$ is the *ensemble ambiguity* measured as the weighted average of the squared differences in the predictions of the base networks and the ensemble. The ambiguity $\bar{A}$ that they used as a way to measure the disagreement, is common when disagreement among numeric predictors such as neural networks is measured.

A similar dependency derived by Tumer and Ghosh [41] for classifiers that predict the a posteriori probabilities of the output classes is:

$$E = \frac{1 + (S - 1) \cdot \sum_{i=1}^L P_i \delta_i}{S} \cdot \bar{E}, \quad (2)$$

where E is the ensemble generalization error (beyond the optimal Bayesian error), S is the size of the ensemble, P_i is the prior probability of class i , δ_i is the *average correlation factor* among the classifiers in the prediction of the a posteriori probabilities of class i , and $\bar{E}$ is the average error of the base classifier. Here the disagreement is measured as the correlation in the predictions of the a posteriori probabilities.

While these theoretical results hold true for regression systems and classifiers that predict a posteriori probabilities, no similar dependencies are known for the more common case of classifiers with crisp predictions, which is the focus of the paper. However, there are a number of approaches to quantify ensemble diversity for the case of crisp classification, which we consider in Section 4.

2.2 Techniques for the generation of base classifiers in ensembles

The task of using an ensemble of models can be broken down into two basic questions: (1) what set of learned models should be generated?; and (2) how should the predictions of the learned models be integrated? [24]. To generate a set of accurate and diverse learned models, several approaches have been tried.

One way of generating a diverse set of models is to use learning algorithms with heterogeneous representations and search biases [24], such as decision trees, neural networks, instance-based learning, etc.

Another approach is to use models with homogeneous representations that differ in their method of search or in the data on which they are trained. This approach includes several techniques for generating base models, such as learning base models from different subsets of the training data. For example, two well-known ensemble methods of this type are bagging and boosting [31].

The base models with homogeneous representations may be binary classifiers that are integrated to implement a multiclass learner (i.e., where the number of class labels is greater than 2). Each classifier in such an ensemble is learnt to distinguish one class label from the others. For example, Dietterich and Bakiri [11] map each class label onto a bit string prior to learning. Bit strings for class labels are designed to be well separated, thus serving as error-correcting output codes (ECOC). An off-the-shelf system for learning binary classifications (e.g., 0 or 1) can be used to build multiple classifiers, one for

each bit in the output code. An instance is classified by predicting each bit of its output code (i.e., label), and then classifying the instance as the label with the “closest” matching output code.

Also, natural randomisation in the process of model search (e.g., random weight setting in the backpropagation algorithm for training neural networks) can be used to build different models with homogeneous representation. The randomisation can also be injected artificially. For example, in [16] a randomised decision tree induction algorithm, which generates different decision trees every time it is run, was used for that purpose.

Another way for building models with homogeneous representations, which proved to be effective, is the use of different subsets of features for each model. For example, in [29] base classifiers are built on different feature subsets, where each feature subset includes features relevant for distinguishing one class label from the others (the number of base classifiers is equal to the number of classes). Finding a set of feature subsets for constructing an ensemble of accurate and diverse base models is also known as ensemble feature selection [27].

Sometimes, a combination of the techniques considered above can be useful in order to provide the desired characteristics of the generated models. For example, a combination of boosting and wagging (which is a kind of bagging technique) is considered by Webb [42].

In addition to these general-purpose methods for generating a diverse ensemble of models, there are learning algorithm-specific techniques. For example, Opitz and Shavlik [28] employ a genetic algorithm in backpropagation to search for a good population of neural network classifiers.

Ensemble feature selection is the focus of this paper, and in next section we consider search strategies used in our experiments.

2.3 Search strategies for ensemble feature selection

In this section, we consider four different search strategies for ensemble feature selection: (1) Hill Climbing (HC); (2) a Genetic Algorithm for ensemble feature selection (GA); (3) Ensemble Forward Sequential Selection (EFSS); and (4) Ensemble Backward Sequential Selection (EBSS).

The use of a hill-climbing search as a local-search wrapper-based approach has been shown to be effective for a single feature subset selection [18]. The Hill Climbing (HC) ensemble feature selection strategy, which we use in this research, proposed in [8], is composed of two major phases: (1) construction of the initial ensemble by random subspacing; and (2) iterative refinement of the ensemble members with sequential-mutation hill climbing. Initial feature subsets are constructed using the random subspace method. Then, the initial ensemble is formed. Further, an iterative refinement of the ensemble members is used to improve the accuracy and diversity of the base classifiers. The iterative

refinement is based on a hill-climbing search. For all the feature subsets, an attempt is made to switch (include or delete) each feature. If the resulting feature subset produces better performance on the validation set (the fitness function returns a better value for it), that change is kept. This process is continued until no further improvements are possible. Normally, no more than 4 passes are necessary.

The use of genetic search has also been an important direction in the feature selection research. Genetic algorithms have been shown to be effective global optimization techniques in feature subset selection. The use of genetic algorithms for ensemble feature selection was first proposed in [27]. The Genetic Algorithm for ensemble feature selection (GA) strategy [27] begins, as HC, with creating an initial population of classifiers where each classifier is generated by randomly selecting a different subset of features. Then, new candidate classifiers are continually produced by using the genetic operators of crossover and mutation on the feature subsets. After producing a certain number of individuals the process continues with selecting a new subset of candidates by selecting the members randomly with a probability proportional to fitness (it is known as roulette-wheel selection). The process of producing new classifiers and selecting a subset of them (a generation) continues a number of times, known as the number of generations. After a predefined number of generations, the fittest individuals make up the population, which comprises the ensemble [27]. In our implementation, the representation of each individual (a feature subset) is simply a constant-length string of bits, where each bit corresponds to a particular feature. The crossover operator uses uniform crossover, in which each feature of the two children takes randomly a value from one of the parents. The feature subsets of two individuals in the current population are chosen randomly with a probability proportional to $(1+fitness)$. The mutation operator randomly toggles a percentage of bits in an individual.

EFSS and EBSS are sequential feature selection strategies, which add or delete features using a hill-climbing procedure, and have polynomial complexity. The most frequently studied variants of plain sequential feature selections algorithms (which select a single feature subset) are forward and backward sequential selection, FSS and BSS [1]. FSS begins with zero attributes, evaluates all feature subsets with exactly one feature, and selects the one with the best performance. It then adds to this subset the feature that yields the best performance for subsets of the next larger size. The cycle repeats until no improvement is obtained from extending the current subset. BSS instead begins with all features and repeatedly removes a feature whose removal yields the maximal performance improvement [1]. EFSS and EBSS iteratively apply FSS or BSS to form each of the base classifiers using a predefined fitness function.

EFSS and EBSS have polynomial complexity with regard to the number of features: $O(S \cdot F \cdot F')$, where S is the number of base classifiers, F is the total number of features, and F' is the number of

features included or deleted on average in an FSS or BSS search. HC has similar polynomial complexity $O(S \cdot F \cdot N_{passes})$, where N_{passes} is the average number of passes through the feature subsets in HC until there is some improvement (usually no more than 4). The complexity of GEFS does not depend on the number of features, and is $O(S' \cdot N_{gen})$, where S' is the number of individuals (feature subsets) in one generation, and N_{gen} is the number of generations.

In our experiments, on average, each of the strategies looks through about 1000 feature subsets (given that the number of base classifiers is 25, the average number of features is 10, the average percentage included or deleted features in EFSS and EBSS is 40%, the number of passes in HC is 4, the number of individuals in a population in the GA is 100, and the number of generations is 10).

Comparative experiments with these four strategies on a collection of data sets from the medical field of acute abdominal pain classification were considered in [38]. The best search strategy in that context was EFSS (it was the best for every data set considered), generating more diverse ensembles with more compact base classifiers. EFSS generated extremely compact base classifiers, including from 9% to 13% features on average (less than 3 features). The results with genetic search were found to be disappointing, not being improved with generations.

In comparison with [38] we have made a number of changes in the genetic search-based strategy. In our new version considered in this paper no full feature sets are allowed in random subsampling nor may the crossover operator produce a full feature subset. Individuals for crossover are selected randomly proportional to $\log(1+fitness)$ instead of just fitness, which adds more diversity into the new population. The generation of children identical to their parents is prohibited in the crossover operator – if a child is the same as one of its parents, the mutation operator is applied to it. To provide a better diversity in the length of the feature subsets and the population in general, we use two different mutation operators, one of which always adds features randomly with a given probability, and the other – deletes features. Each operator is applied to exactly a half (25 of 50) of the individuals being mutated. Our pilot studies have shown that these minor changes help us to significantly improve the work of GA so that it does not converge after a single generation already to a local extremum (as it was in [27, 38]).

Parameter settings for our implementation of the genetic search in GA include a mutation rate of 50% (as proposed in [27]), a population size of 25, a search length of 100 feature subsets, of which 50 are offsprings of the current population of 25 classifiers generated with the crossover operator, and 50 are mutated offsprings. 10 generations of individuals were produced, as our pilot studies have shown that in most cases, with this configuration, the ensemble accuracy does not improve after 10 generations, due to overfitting the training data.

3 Techniques for integration of an ensemble of models

Brodley and Lane [4] have shown that simply increasing coverage of an ensemble through diversity is not enough to insure increased prediction accuracy. If the integration method does not utilize the coverage, then no benefit arises from integrating multiple classifiers. Thus, the diversity and coverage of an ensemble are not in themselves sufficient conditions for ensemble accuracy. It is also important for ensemble accuracy to have a good integration method that will utilize the diversity of the base models.

The challenging problem of integration is to decide which one(s) of the classifiers to rely on or how to combine the results produced by the base classifiers. Techniques using two basic approaches have been suggested as a solution to the integration problem: (1) a *combination approach*, where the base classifiers produce their classifications and the final classification is composed using them; and (2) a *selection approach*, where one of the classifiers is selected and the final classification is the result produced by it.

Several effective techniques for the *combination* of classifiers have been proposed. One of the most popular and simplest techniques used to combine the results of the base classifiers, is simple voting (also called majority voting and select all majority (SAM)) [2]. In the voting technique, the classification of each base classifier is considered as an equally weighted vote for that particular classification. The classification that receives the biggest number of votes is selected as the final classification (ties are solved arbitrarily). Often, weighted voting is used: each vote receives a weight, which is usually proportional to the estimated generalization performance of the corresponding classifier. Weighted Voting (WV) works usually much better than simple majority voting [2].

More sophisticated combination techniques include the SCANN method based on the correspondence analysis and using the nearest neighbor search in the correspondence analysis results [24, 26]; and techniques to combine minimal nearest neighbor classifiers within the stacked generalization framework [35]. Two effective classifier combination techniques based on stacked generalization called “arbiter” and “combiner” were presented by Chan [5]. Hierarchical classifier combination has also been considered. Experimental results of Chan and Stolfo [5, 6] showed that the hierarchical (multi-level) combination approach, where the dataset was distributed among a number of sites, was often able to sustain the same level of accuracy as a global classifier trained on the entire dataset.

A number of *selection* techniques have also been proposed to solve the integration problem. One of the most popular and simplest selection techniques is the cross-validation majority (CVM, we call it simply Static Selection, SS, in our experiments) [33]. In CVM, the cross-validation accuracy for each

base classifier is estimated using the training set, and then the classifier with the highest accuracy is selected (ties are solved using voting). More sophisticated selection approaches include estimation of the local accuracy of the base classifiers by considering errors made in instances with similar predictions [25], learning a number of meta-level classifiers (“referees”) that predict whether the corresponding base classifiers are correct or not for new instances (each “referee” is a C4.5 tree that recognizes two classes) [20]. Todorovski and Dzeroski [37] trained a meta-level decision tree, which dynamically selected a base model to be applied to the considered instance, using the level of confidence of the base models in correctly classifying the instance.

The approaches to classifier selection can be divided into two subsets: *static* and *dynamic* selection. The static approaches propose one “best” method for the whole data space, while the dynamic approaches take into account each new instance to be classified and its neighbourhood only. The CVM is an example of the static approach, while the other selection techniques considered above are examples of the dynamic approach.

Techniques for combining classifiers can be static or dynamic as well. For example, widely used weighted voting [2] is a static approach. The weights for each base classifier’s vote do not depend on the instance to be classified. In contrast, the reliability-based weighted voting (RBWV) introduced in [7] is a dynamic voting approach. It uses classifier-dependent estimation of the reliability of predictions for each particular instance. Usually, better results can be achieved if the classifier integration is done dynamically taking into account the characteristics of each new instance.

We consider in our experiments three dynamic approaches based on the local accuracy estimates: Dynamic Selection (DS) [30], Dynamic Voting (DV) [30], and Dynamic Voting with Selection (DVS) [40]. All these are based on the same local accuracy estimates. DS simply selects a classifier with the least predicted local classification error, as was also proposed in [14]. In DV, each base classifier receives a weight that is proportional to the estimated local accuracy of the base classifier, and the final classification is produced by combining the votes of each classifier with their weights. In DVS, the base classifiers with highest local classification errors are discarded (the classifiers with errors that fall into the upper half of the error interval of the base classifiers) and locally weighted voting (DV) is applied to the remaining base classifiers.

4 Diversities

There are a number of ways to quantify ensemble diversity. In this section we consider seven different measures of ensemble diversity, five of which are pairwise as they are able to measure diversity in predictions of a pair of classifiers, and the total ensemble diversity is the average of all the

classifier pairs in the ensemble (plain disagreement, fail/non-fail disagreement, the Q statistic, the correlation coefficient, and the kappa statistic), and the other two are non-pairwise as they measure diversity in predictions of the whole ensemble only (entropy and ambiguity). In our experiments in this paper, we compare the use of the pairwise measures as the guiding diversity in the ensemble training process (as a component of the fitness function (1)), and consider the correlation of the total average ensemble diversity for all the seven measures with the difference between the ensemble accuracy and the average base classifier accuracy.

4.1 The plain disagreement measure

The plain disagreement measure of diversity is, probably, the most commonly used measure for diversity in the ensembles of classifiers with crisp predictions. For example, in [17] it was used for measuring the diversity of decision forests, and its correlation with the forests' accuracy. In [39, 43] it was used as a component of the fitness function guiding the process of ensemble construction.

For two classifiers i and j , the plain disagreement is equal to the ratio between the number of instances on which the classifiers make different predictions, to the total number of instances in question:

$$div_plain_{i,j} = \frac{1}{N} \sum_{k=1}^N \text{Diff}(C_i(\mathbf{x}_k), C_j(\mathbf{x}_k)), \quad (3)$$

where N is the number of instances in the data set, $C_i(\mathbf{x}_k)$ is the class assigned by classifier i to instance k , and $\text{Diff}(a,b)=0$, if $a=b$, otherwise $\text{Diff}(a,b)=1$.

The plain disagreement varies from 0 to 1. This measure is equal to 0, when the classifiers return the same classes for each instance, and it is equal to 1 when the predictions are always different.

4.2 The fail/non-fail disagreement measure

The fail/non-fail disagreement was defined in [36] as the percentage of test instances for which the classifiers make different predictions but for which one of them is correct:

$$div_dis_{i,j} = \frac{N^{01} + N^{10}}{N^{11} + N^{10} + N^{01} + N^{00}}, \quad (4)$$

where N^{ab} is the number of instances in the data set, classified correctly ($a=1$) or incorrectly ($a=0$) by the classifier i , and correctly ($b=1$) or incorrectly ($b=0$) by the classifier j . The denominator in (4) is equal to the total number of instances N . (4) is equal to (3) for binary classification problems, where the

number of classes is 2. It can be also shown that $div_dis_{i,j} \leq div_plain_{i,j}$, as the instances contributing to this disagreement measure form a subset of instances contributing to the plain disagreement.

The fail/non-fail disagreement varies from 0 to 1. This measure is equal to 0, when the classifiers return the same classes for each instance, or different but incorrect classes, and it is equal to 1 when the predictions are always different and one of them is correct.

4.3 The Q statistic

This measure is based on Yule's Q statistic used to assess the similarity of two classifiers' outputs [22]:

$$div_Q_{i,j} = \frac{N^{11}N^{00} - N^{01}N^{10}}{N^{11}N^{00} + N^{01}N^{10}}, \quad (5)$$

where N^{ab} has the same meaning as in (4). For statistically independent classifiers, the expected value of Q is 0. Q varies between -1 and 1 . Classifiers that tend to recognize the same objects correctly will have positive values of Q , and those which commit errors on different objects will render Q negative [22]. In the case of undefined value with division by zero, we assume the diversity is minimal, equal to 1.

In our experiments with diversity as a component of the fitness function we normalize this measure to vary from 0 to 1, where 1 corresponds to the maximum of diversity:

$$div_Q^* = \frac{1 - div_Q}{2}. \quad (6)$$

In [22], after comparative experiments on the UCI Breast cancer Wisconsin data set, the Phoneme recognition and the Cone-torus data sets, and two experiments with emulated ensembles (artificially generating possible cases of the base classifiers' outputs), Q was recommended as the best measure for the purposes of developing committees that minimize error, taking into account the experimental results, its simplicity and comprehensibility (or the ease of interpretation). In our experiments, we show that the question of the superiority of a particular measure depends on the context of the use of diversity, and on the data being processed.

One problem, which we have noticed with this measure in our pilot studies, was its insensitivity on small data sets. It is quite an often the case for a small number of instances that N^{00} is equal to 0. Q in this case is equal to -1 (maximal diversity) no matter how big the values of N^{01} and N^{10} are, which is often not a good reflection of the real diversity in classifiers' outputs.

4.4 The correlation coefficient

The correlation between the outputs of two classifiers i and j can be measured as [22]:

$$div_corr_{i,j} = \frac{N^{11}N^{00} - N^{01}N^{10}}{\sqrt{(N^{11} + N^{10})(N^{01} + N^{00})(N^{11} + N^{01})(N^{10} + N^{00})}}, \quad (7)$$

where N^{ab} have the same meaning as in (4) and (5).

The numerator in (7) is the same as in (5), and for any two classifiers i and j , $div_corr_{i,j}$ and $div_Q_{i,j}$ have the same sign, and it can be proven that $|div_corr_{i,j}| \leq |div_Q_{i,j}|$ [22]. We normalize this measure to vary from 0 to 1 in the same way we do it for Q (5).

This measure, as well as the fail/non-fail disagreement and the Q statistic were considered among the group of 10 measures in the comparative experiments in [22].

4.5 The kappa degree-of-agreement statistic

Let N_{ij} be the number of instances in the data set, recognized as class i by the first classifier and as class j by the second one, N_{i*} is the number of instances recognized as i by the first classifier, and N_{*i} is the number of instances recognized as i by the second classifier. Define then Θ_1 and Θ_2 as

$$\Theta_1 = \frac{\sum_{i=1}^l N_{ii}}{N}, \text{ and } \Theta_2 = \sum_{i=1}^l \left(\frac{N_{i*}}{N} \cdot \frac{N_{*i}}{N} \right), \quad (8)$$

where N is the total number of instances. Θ_1 estimates the probability that the two classifiers agree, and Θ_2 is a correction term for Θ_1 , which estimates the probability that the two classifiers agree simply by chance (in the case where each classifier chooses to assign a class label randomly). The pairwise diversity $div_kappa_{i,j}$ is defined as follows [9]:

$$div_kappa_{i,j} = \frac{\Theta_1 - \Theta_2}{1 - \Theta_2}. \quad (9)$$

$Kappa$ is equal to 0 when the agreement of the two classifiers equals to that expected by chance, and $kappa$ is equal to 1 when the two classifiers agree on every example. Negative values occur when agreement is less than expected by chance – that is, there is systematic disagreement between the classifiers [9]. $Kappa$ is able to track negative correlations in a similar manner to Q and correlation. We normalize this measure to vary from 0 to 1 in the same way we do for Q and correlation (5), (7).

Dietterich [9] used this measure in scatter plots called “ κ -error diagrams”, where kappa was plotted against mean accuracy of the classifier pair. κ -error diagrams, introduced in [23] is a useful tool for visualising ensembles.

4.6 Entropy

A non-pairwise measure of diversity, associated with a conditional-entropy error measure, is based on the concept of entropy [8]:

$$div_ent = \frac{1}{N} \sum_{i=1}^N \sum_{k=1}^l -\frac{N_k^i}{S} \cdot \log\left(\frac{N_k^i}{S}\right), \quad (10)$$

where N is the number of instances in the data set, S is the number of base classifiers, l is the number of classes, and N_k^i is the number of base classifiers that assign instance i to class k . To keep this measure of diversity within the range $[0,1]$ the logarithm should be taken to the base l .

This measure was evaluated on a medical prediction problem and was shown to predict the accuracy of the ensemble well [8]. It was also shown that the entropy measure of diversity has the added advantage that it models the change in diversity with the size of the ensemble.

While this may be a useful measure of diversity, it does not allow us to gauge the contribution of an individual classifier to diversity because it is not a pairwise measure [43]. Thus, we shall use this total measure of diversity only in our experiments on the correlation between the total ensemble diversity and the difference in ensemble accuracy and the base classifier accuracy.

4.7 A variance-based measure: Ambiguity

This non-pairwise measure of diversity is associated with the variance-based measure of diversity proposed for regression problems in [21], called *ambiguity*. This diversity has been proven to have a direct relation with the ensemble error and this motivated us to use an associated diversity measure for classification also. The classification task can be decomposed into l regression tasks, where l is the number of classes. The output in the regression tasks will be the class membership of the instance (binary output 0/1 in the case of crisp classification considered in this paper). The diversity of the classification ensemble can then be calculated as the average ambiguity over these pseudo-regression tasks for each of the instances:

$$div_amb = \frac{1}{lN} \sum_{i=1}^l \sum_{j=1}^N Ambiguity_{i,j} = \frac{1}{lNS} \sum_{i=1}^l \sum_{j=1}^N \sum_{k=1}^S \left(\text{Is}(C_k(\mathbf{x}_j) = i) - \frac{N_i^j}{S} \right)^2, \quad (11)$$

where l is the number of classes, N is the number of instances, S is the number of base classifiers, N_i^j is the number of base classifiers that assign instance j to class i , $C_k(\mathbf{x}_j)$ is the class assigned by classifier k to instance j , and $\text{Is}()$ is a truth predicate.

As with entropy, this is a non-pairwise measure, and we shall use it only in our experiments on the correlation between the total ensemble diversity and the difference in the ensemble accuracy and the base classifier accuracy.

5 Experimental investigations

In this section, our experiments with the seven measures of diversity in ensemble feature selection are presented. First, the experimental setting is described, then the results of the experiments are presented. The experiments are conducted on 21 data sets taken from the UCI machine learning repository [3]. These data sets include real-world and synthetic problems, vary in characteristics, and were previously investigated by other researchers.

The main characteristics of the 21 data sets are presented in Table 1. The table includes the name of a data set, the number of instances included in the data set, the number of different classes of instances in the data set, and the numbers of different kinds of features included in the instances of the data set.

In [32] the use of the data sets from the UCI repository was strongly criticized. Any new experiments on the UCI data sets run the risk of finding “significant” results that are no more than statistical accidents. However, we do believe that nevertheless, the UCI data sets provide useful benchmark estimates that can be of great help in experimental analysis and comparisons of learning methods, even though these results must be looked at carefully before final conclusions are made.

Table 1. Data sets and their characteristics

Data set	Instances	Classes	Features	
			Categorical	Numerical
Balance	625	3	0	4
Breast Cancer Ljubljana	286	2	9	0
Car	1728	4	6	0
Pima Indians Diabetes	768	2	0	8
Glass Recognition	214	6	0	9
Heart Disease	270	2	0	13
Ionosphere	351	2	0	34
Iris Plants	150	3	0	4
LED	300	10	7	0

LED17	300	10	24	0
Liver Disorders	345	2	0	6
Lymphography	148	4	15	3
MONK-1	432	2	6	0
MONK-2	432	2	6	0
MONK-3	432	2	6	0
Soybean	47	4	0	35
Thyroid	215	3	0	5
Tic-Tac-Toe Endgame	958	2	9	0
Vehicle	846	4	0	18
Voting	435	2	16	0
Zoo	101	7	16	0

5.1 Experimental setting

For our experiments, we used an updated version of the setting presented in [39] to test the EFS_SBC algorithm (Ensemble Feature Selection with the Simple Bayesian Classification). We extended the EFS_SBC setting with an implementation of three new search strategies besides the existing HC: GA, EFSS, and EBSS (Section 2.3), and with an implementation of six new measures of diversity besides the existing plain disagreement: the fail/non-fail disagreement, the Q statistic, the correlation coefficient, the $kappa$ statistic, entropy and ambiguity (Section 4).

We used the simple Bayesian classification as the base classifiers in the ensembles. It has been recently shown experimentally and theoretically that the simple Bayes can be optimal even when the “naïve” feature-independence assumption is violated by a wide margin [12]. Second, when the simple Bayes is applied to the subproblems of lower dimensionalities as in random subsampling, the error bias of the Bayesian probability estimates caused by the feature-independence assumption becomes smaller. It also can easily handle missing feature values of a learning instance allowing the other feature values still to contribute. Besides, it has advantages in terms of simplicity, learning speed, classification speed, and storage space, which made it possible to conduct all the experiments within reasonable time. It was shown [39] that only one “global” contingency table is needed for the whole ensemble when the simple Bayes is employed in ensemble feature selection. We believe that the results presented in this paper do not depend significantly on the learning algorithm used and would be similar for most known learning algorithms.

To estimate the ensemble accuracy after ensemble feature selection, we have used random-sampling cross validation. 70 test runs of EFS_SBC are made for each search strategy, for each diversity measure, and on each data set. In each test run the data set is first split into the training set TrS, the validation set VS, and the test set TS by stratified random sampling. In stratified random sampling (sometimes called proportional random sampling), a simple random sample from instances of each

class is taken, so that the class distributions of instances in the resulting sets are approximately the same as in the initial data set. It was shown that such sampling often gives better accuracy estimates than simple random sampling [18]. Each time 60 percent of instances are assigned to the training set. The remaining 40 percent of instances of the data set are divided into two sets of approximately equal size (VS and TS). The validation set VS is used in the ensemble refinement to estimate the accuracy and diversity guiding the search process. The test set TS is used for the final estimation of the ensemble accuracy. We have used the same division of the data into the training, validation and test sets for each search strategy and guiding diversity to avoid unnecessary variance and provide better comparison.

The ensemble size S was selected to be 25. It has been shown that for many ensembles, the biggest gain in accuracy is achieved already with this number of base classifiers [2].

We experimented with seven different values of the diversity coefficient α : 0, 0.25, 0.5, 1, 2, 4, and 8. At each run of the algorithm, we collect accuracies for the five types of integration of the base classifiers (Section 3): Static Selection (SS), Weighted Voting (WV), Dynamic Selection (DS), Dynamic Voting (DV), and Dynamic Voting with Selection (DVS). In the dynamic integration strategies DS, DV, and DVS, the number of nearest neighbors (k) for the local accuracy estimates was pre-selected from the set of seven values: 1, 3, 7, 15, 31, 63, 127 ($2^n - 1$, $n = 1, \dots, 7$), for each data set separately, if the number of instances in the training set permitted. Such values were chosen, as the nearest-neighbor procedure in the dynamic integration is distance-weighted. The distances from the test instance to the neighboring training instances were computed and used to calculate the estimates of local accuracies. As it was shown in [14], this makes the dynamic integration less sensitive to the choice of the number of neighbors, but still the local accuracy estimates are more sensitive to the small number of neighbors. Heterogeneous Euclidean-Overlap Metric (HEOM) [30] was used for calculation of the distances. In the dynamic integration, we used 10-fold cross validation for building the cross-validation history for the estimation of the local accuracies of the base classifiers.

To select the best α and k , we conducted a separate series of experiments using the wrapper approach for each combination of a search strategy, guiding diversity, integration method, and data set. After, the experiments were repeated with the pre-selected values of α and k .

Besides the test-set classification accuracies of the base classifiers and the ensembles, we collected other characteristics as the ensemble accuracies on the validation set (to measure overfitting), the total ensemble diversity (using the seven measures presented in Section 4), the ensemble coverage, and the average relative number of features in the base classifiers. For the GA strategy all the ensemble characteristics were collected after 1, 5, and 10 generations.

The test environment was implemented within the MLC++ framework (the machine learning library in C++) [19]. A multiplicative factor of 1 was used for the Laplace correction in simple Bayes as in [12]. Numeric features were discretized into ten equal-length intervals (or one per observed value, whichever was less), as it was done in [12]. Although this approach was found to be slightly less accurate than more sophisticated ones, it has the advantage of simplicity, and is sufficient for comparing different ensembles of simple Bayesian classifiers with each other, and with the “global” simple Bayesian classifier. The use of more sophisticated discretization approaches could lead to better classification accuracies for the base classifiers and ensembles, but should not influence the results of the comparison.

5.2 Correlation between the total ensemble diversity and the difference between the ensemble accuracy and the average base classifier accuracy

First, we measured correlation between the total ensemble diversity and the difference between the ensemble accuracy and the average base classifier accuracy. To measure the correlation we used the commonly used Pearson’s linear correlation coefficient r and a non-parametric counterpart to it - Spearman’s rank correlation coefficient (RCC).

RCC is a nonparametric (distribution-free) rank statistic proposed by Spearman in 1904 as a measure of the strength of the associations between two variables. It is a measure of monotone association that is used when the distribution of the data make Pearson’s correlation coefficient undesirable or misleading. Kuncheva and Whitaker [22] have used RCC in similar experiments, motivating the choice by the fact that the relationships were not linear. RCC is based on the difference in ranks of the corresponding variables x and y (N here is the number of observations):

$$RCC = 1 - 6 \cdot \sum_{i=1}^N \frac{(Rank(x_i) - Rank(y_i))^2}{N(N^2 - 1)}. \quad (12)$$

In our case, N is equal to 70, corresponding to 70 different measures of the total ensemble diversity and the difference in the ensemble accuracy and the average base classifier accuracy for the 70 cross-validation runs for each data set. We have measured these correlations for the four search strategies, EBSS, EFSS, GA, and HC, and for simple random subsampling (RS). In the search strategies we have used all the five pairwise measures of diversity considered in Section 4.

Our first finding was that the correlation depends greatly on the data set (varying from -0.374 to 0.997) and does not depend on the search strategy used (the difference in the corresponding correlation

coefficients was not more than 0.017 for every data set), so we continue our analysis only for the ensembles built with random subsampling.

In Table 2, Pearson’s correlation coefficient (r) is presented that quantifies the correlation between the total ensemble diversity and the difference between the ensemble accuracy and the average base classifier accuracy for the ensembles built with random subsampling. All the seven diversity measures (Section 4) are considered, corresponding to different columns of the table: (1) the plain disagreement, *div_plain*; (2) the fail/non-fail disagreement, *div_dis*; (3) the Q statistic, *div_Q*; (4) the correlation coefficient, *div_corr*; (5) the $kappa$ statistic, *div_kappa*; (6) entropy, *div_ent*; and (7) ambiguity, *div_amb*. We measured the correlations for three integration methods (DVS, which is the strongest of the dynamic integration methods, Weighted Voting, WV, and plain non-weighted Voting), presented in the lines of the table. Each cell corresponding to a measure of diversity and an integration method contains three values: average, maximal and minimal values of the 21 correlation coefficients, calculated for the 21 data sets. Table 3 has the same structure as Table 2 and presents Spearman’s rank correlation coefficients (RCC).

Table 2. Pearson’s correlation coefficient (r) for the total ensemble diversity and the difference between the ensemble accuracy and the average base classifier accuracy (average, maximal and minimal values)

r	<i>div_plain</i>	<i>div_dis</i>	<i>div_Q</i>	<i>div_corr</i>	<i>div_kappa</i>	<i>div_ent</i>	<i>div_amb</i>
DVS	0.534	0.534	0.383	0.436	0.492	0.527	0.534
	0.997	0.994	0.702	0.810	0.991	0.979	0.997
	-0.021	-0.011	-0.121	0.007	-0.163	0.021	-0.021
WV	0.459	0.477	0.380	0.435	0.398	0.462	0.459
	0.997	0.994	0.791	0.910	0.991	0.979	0.997
	0.074	0.108	-0.019	0.040	-0.206	0.108	0.075
Voting	0.363	0.404	0.346	0.382	0.292	0.373	0.362
	0.997	0.994	0.791	0.910	0.991	0.979	0.997
	-0.312	-0.312	-0.257	-0.360	-0.369	-0.297	-0.312

Table 3. Spearman’s rank correlation coefficient (RCC) for the total ensemble diversity and the difference between the ensemble accuracy and the average base classifier accuracy (average, maximal and minimal values)

RCC	<i>div_plain</i>	<i>div_dis</i>	<i>div_Q</i>	<i>div_corr</i>	<i>div_kappa</i>	<i>div_ent</i>	<i>div_amb</i>
DVS	0.515	0.516	0.365	0.424	0.479	0.505	0.514
	0.992	0.990	0.679	0.791	0.984	0.970	0.992
	-0.033	-0.024	-0.224	0.010	-0.098	0.009	-0.030
WV	0.448	0.469	0.373	0.443	0.398	0.449	0.447
	0.992	0.990	0.682	0.938	0.984	0.970	0.992
	0.049	0.079	-0.083	0.002	-0.235	0.107	0.051

	0.355	0.399	0.345	0.388	0.288	0.363	0.354
Voting	0.992	0.990	0.682	0.938	0.984	0.970	0.992
	-0.274	-0.274	-0.219	-0.343	-0.374	-0.262	-0.276

From Tables 2 and 3, it can be seen that the r and RCC coefficients uncover the same dependencies for the diversities in this context. In fact, for the vast majority of data sets, the difference between the r and RCC was at most only 0.05 (for all the diversities), and more than 0.05 (but no more than 0.11) – only for data sets with small negative correlations. The correlations change greatly with the data sets; this can be seen from the minimal and maximal values presented. Naturally, for some data sets, where ensembles are of little use, the correlations are low or even negative.

For the DVS integration method (the strongest integration method), the best correlations are shown by *div_plain*, *div_dis*, *div_ent*, and *div_amb*. *Div_kappa* is very close to the best. *Div_Q* and *div_corr* behave in a similar way, which reflects the mentioned similarity in their formulae. *Div_Q* has surprisingly the worst average correlation with DVS. However, this correlation does not significantly decrease with the change of the integration method to WV and Voting, as it happens with all the other diversities (!) besides *div_corr*. The decrease in the correlations with the change of the integration method from DVS to WV, and from WV to Voting can be explained by the fact that DVS better utilizes the ensemble diversity in comparison with WV (as it was shown, for example, in [39]), and WV better utilizes the ensemble diversity than Voting, and, naturally, the correlation coefficients change as well. The fact that *div_Q* and *div_corr* do not decrease with the change of integration method is surprising, and, probably, can be explained by the different way of calculation of the diversity, but this needs further research. *Div_dis* is the best measure of diversity on average in this context.

Div_plain is almost equal to *div_amb*, the difference between them is at most 0.001 only. Possibly, this similarity can be proven theoretically. *Div_amb* could be a good measure of diversity for classifiers that output the a posteriori probability of the output classes – this needs further research.

The line with the maximal correlations corresponds to the Soybean data set. The dependency is almost perfectly linear for this data set for all the diversities besides *div_Q* and *div_corr*. This behaviour, probably, can be explained by the characteristics of the data set (it is the smallest data set including 47 instances only, and it is easy to learn), and needs further research.

5.3 Comparison of the test set accuracy for different strategies, diversity metrics, and integration methods

In Figure 1, the ensemble accuracy is shown for different search strategies including simple random subsampling (RS), and for the five pairwise measures of diversity used within the fitness function (1). The best integration method was selected for each data set, and the results were averaged over all the 21 data sets.

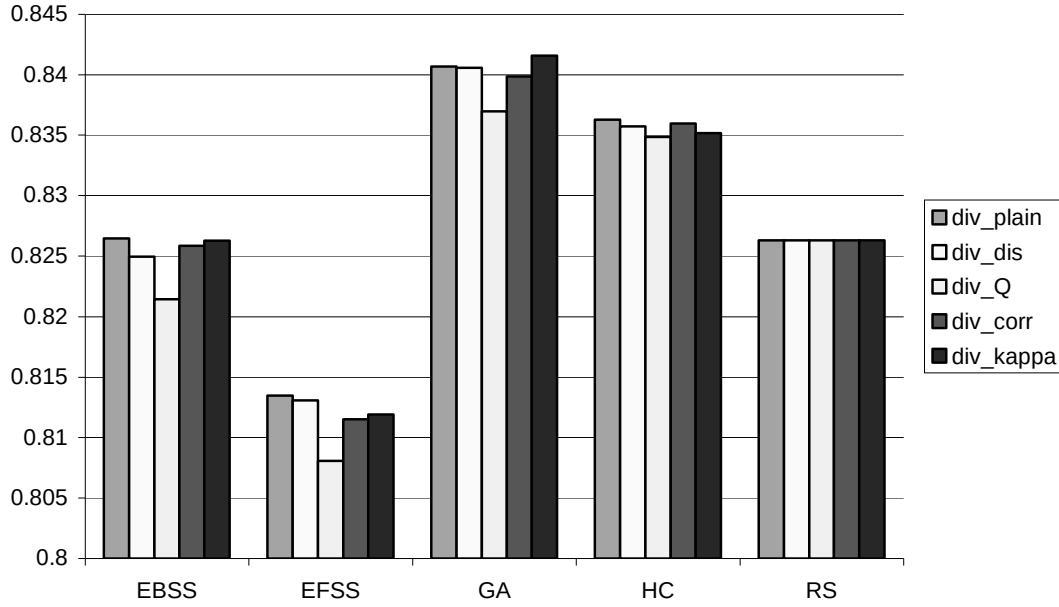


Figure 1. Average test set accuracy for different search strategies and guiding diversities

From Figure 1, one can see that the GA was the best on average, and the HC strategy was the second best. These two strategies are based on random subsampling (RS), which shows very good results on these data sets in itself. RS was better than EFSS for all the diversities, and better or equal than EBSS. Surprisingly, after it was the best for the Acute Abdominal Pain data sets in [38], EFSS was significantly worse than all the other strategies for this collection of data sets. This is more in line with the results of Aha and Bankert [1], which show that, for simple feature selection, backward sequential selection is often better than forward sequential selection as forward selection is not able to include groups of correlated features that do not have high relevance as separate features, only as a group.

There is no significant difference in the use of the guiding diversities. The only two significant dependencies found are: (1) *div_Q* is significantly worse than the other diversities for EBSS, EFSS, and the GA; and (2) *div_kappa* is better than the other diversities for the GA. The first dependency is expected and goes in line with the results presented in Tables 1 and 2. The second dependency is

unexpected and can be explained, probably, by the peculiarities of the genetic search. HC is not that sensitive to the choice of the guiding diversity, as it starts with the already accurate ensemble (RS), and the search is not global as in the GA.

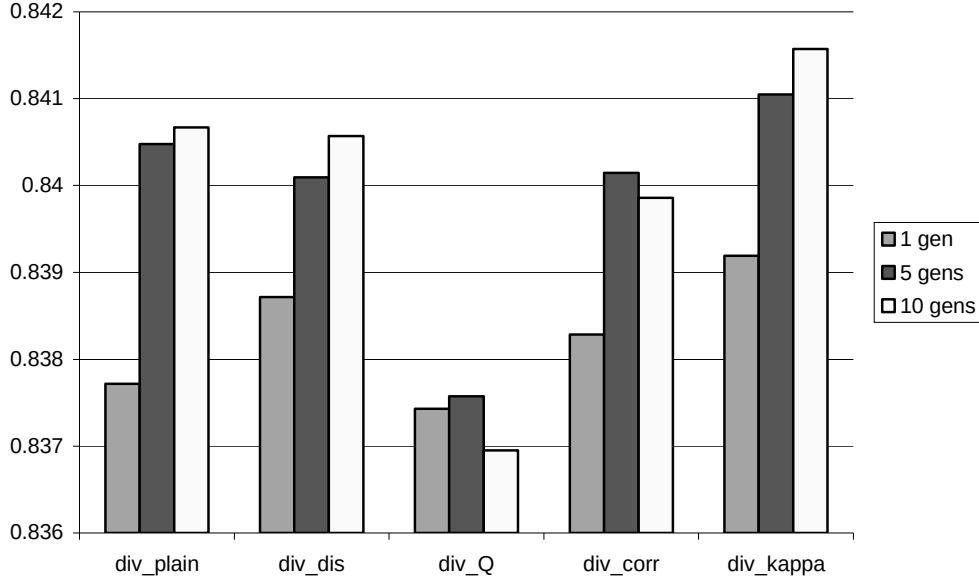


Figure 2. Average test set accuracy for different numbers of generations in the GA search strategy with different guiding diversities

In Figure 2, the average ensemble accuracy is shown for the GA strategy with the five guiding diversities after 1, 5, and 10 generations. From this figure, one can see that ensembles for all the diversities show improvement as the search progresses for the number of generations from 1 to 5. Then, ensembles built with *div_Q* and *div_corr* begin to degenerate, while the other ensembles still improve the average accuracy, though this improvement starts to tail off. This corresponds to the results presented in Tables 1 and 2, where these two measures of diversity have shown the worst average correlation with the difference between the ensemble accuracy and the average base classifier accuracy for DVS. As was also shown in the previous figure, the best diversity for the GA is *div_kappa*.

For each guiding diversity, for each search strategy, and for each data set we have compared the ensemble accuracy of the 70 cross-validation runs using Student’s *t*-test for significance of the difference in two proportions with the level of significance 0.01. The results of this are presented in Table 4. It was shown that the resampled *t*-test and other commonly used significance tests have an unacceptably high probability of detecting a difference in generalization performance when no difference exists (Type 1 error) [32]. However, this is still the most commonly used procedure for comparing learning methods, and it does provide approximate confidence intervals that can be of great help in interpreting experimental comparisons of learning methods.

Each cell of Table 4 compares the diversity metric corresponding to the line against the metric corresponding to the column, and contains win/tie/loss information over the 21 data sets for the four search strategies correspondingly (EBSS, EFSS, GA, and HC), and in total (these numbers are given in gray, bold and italic). For each line and for each column we have also calculated the total numbers, which compare corresponding diversity against all the others, summing up corresponding cells.

Table 4. Student’s *t*-test results comparing guiding diversities for different search strategies on 70 cross-validation runs

Win/tie/loss BSS FSS GA HC <i>total</i>		div_plain	div_dis	div_corr	div_Q	div_kappa	total
div_plain	X		1/20/0	0/21/0	2/19/0	1/19/1	4/79/1
			0/20/1	1/19/1	3/18/0	2/17/2	6/74/4
			0/21/0	0/21/0	3/18/0	0/21/0	3/81/0
			0/21/0	0/21/0	0/21/0	2/19/0	2/82/0
			1/82/1	1/82/1	8/76/0	5/76/3	15/316/5
div_dis	X	0/20/1		0/21/0	1/20/0	1/17/3	2/78/4
		1/20/0		1/20/0	3/18/0	2/17/2	7/75/2
		0/21/0		0/21/0	1/20/0	0/20/1	1/82/1
		0/21/0		0/21/0	0/21/0	2/19/0	2/82/0
		1/82/1		1/83/0	5/79/0	5/73/6	12/317/7
div_corr	X	0/21/0	0/21/0		2/19/0	1/17/3	3/78/3
		1/19/1	0/20/1		2/19/0	3/15/3	6/73/5
		0/21/0	0/21/0		1/20/0	0/19/2	1/81/2
		0/21/0	0/21/0		0/21/0	2/19/0	2/82/0
		1/82/1	0/83/1		5/79/0	6/70/8	13/314/10
div_Q	X	0/19/2	0/20/1	0/19/2		0/16/5	0/74/10
		0/18/3	0/18/3	0/19/2		2/16/3	2/71/11
		0/18/3	0/20/1	0/20/1		0/18/3	0/76/8
		0/21/0	0/21/0	0/21/0		1/19/1	1/82/1
		0/76/8	0/79/5	0/79/5		3/69/12	3/303/30
div_kappa	X	1/19/1	3/17/1	3/17/1	5/16/0		12/69/3
		2/17/2	2/17/2	3/15/3	3/16/2		10/65/9
		0/21/0	1/20/0	2/19/0	3/18/0		6/78/0
		0/19/2	0/19/2	0/19/2	1/19/1		1/76/7
		3/76/5	6/73/5	8/70/6	12/69/3		29/288/19
total	X	1/79/4	4/78/2	3/78/3	10/74/0	3/69/12	
		4/74/6	2/75/7	5/73/6	11/71/2	9/65/10	
		0/81/3	1/82/1	2/81/1	8/76/0	0/78/6	
		0/82/2	0/82/2	0/82/2	1/82/1	7/76/1	
		5/316/15	7/317/12	10/314/13	30/303/3	19/288/29	

This table supports our conclusions from Figure 1. There is no significant difference in the use of the guiding diversities in most cases. However, *Div_Q* is significantly worse than the others for EBSS (10 losses and no wins of the 84 comparisons), for EFSS (11 losses and 2 wins), and for GA (8 losses and no wins). For HC there is no difference in the use of any diversity. *Div_kappa* is the best diversity for

the GA (6 wins and no losses) and it works well for EBSS (12 wins against 3 losses), however, it is very unstable for EFSS (10 wins and 9 losses). The other differences are insignificant.

As the rest of the dependencies presented in this section do not depend on what guiding diversity is used, we consider them only for *div_plain* as the guiding diversity. In Figure 3, the average ensemble accuracies for the five integration methods (SS, WV, DS, DV, and DVS) are shown in combination with the four search strategies (EBSS, EFSS, GA, and HC) and RS.

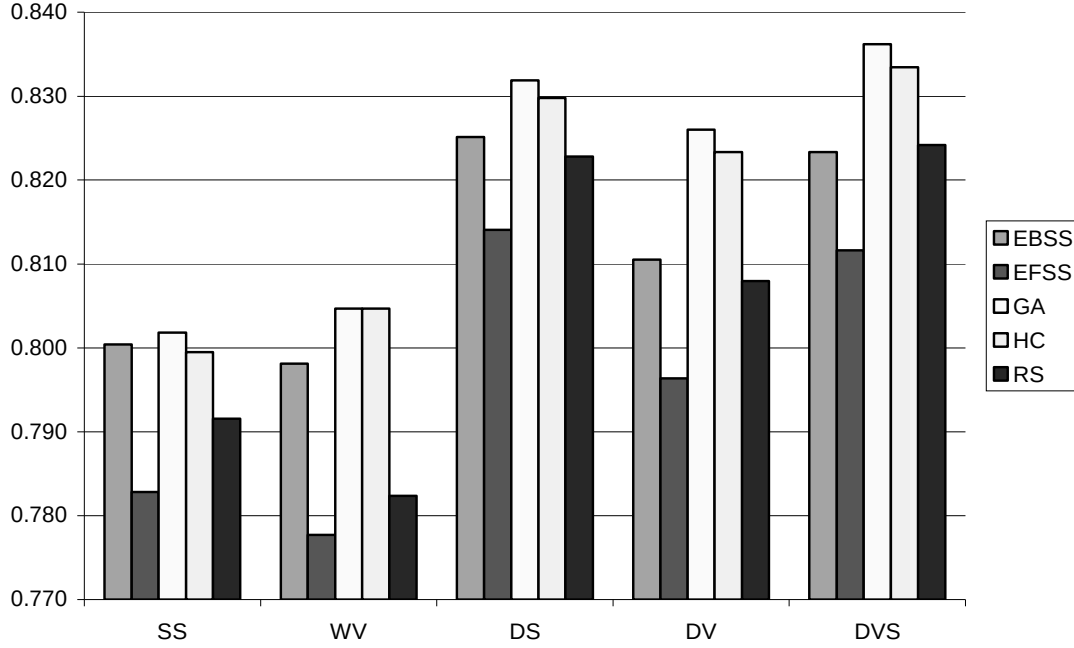


Figure 3. Average test set accuracy for different integration methods with different search strategies

From Figure 3, one can see that the dynamic methods always work better on average than the static methods, and DVS is the best dynamic method, that supports the conclusions presented in [38, 39]. DVS for the GA shows the best accuracy, and DVS with HC is the second best. For each of the dynamic methods, the ranking of the search strategies is the same: GA, HC, EBSS, RS, and EFSS.

5.4 Average test set accuracy for different search strategies and integration methods on data sets with different numbers of features

In Figures 4 and 5, the same information (the average ensemble accuracy for the four integration methods with the four search strategies and RS) as in Figure 3 is shown, but for the two groups of data sets: (1) with less than 9 features (10 data sets), and (2) with more or equal to 9 features (11 data sets), correspondingly.

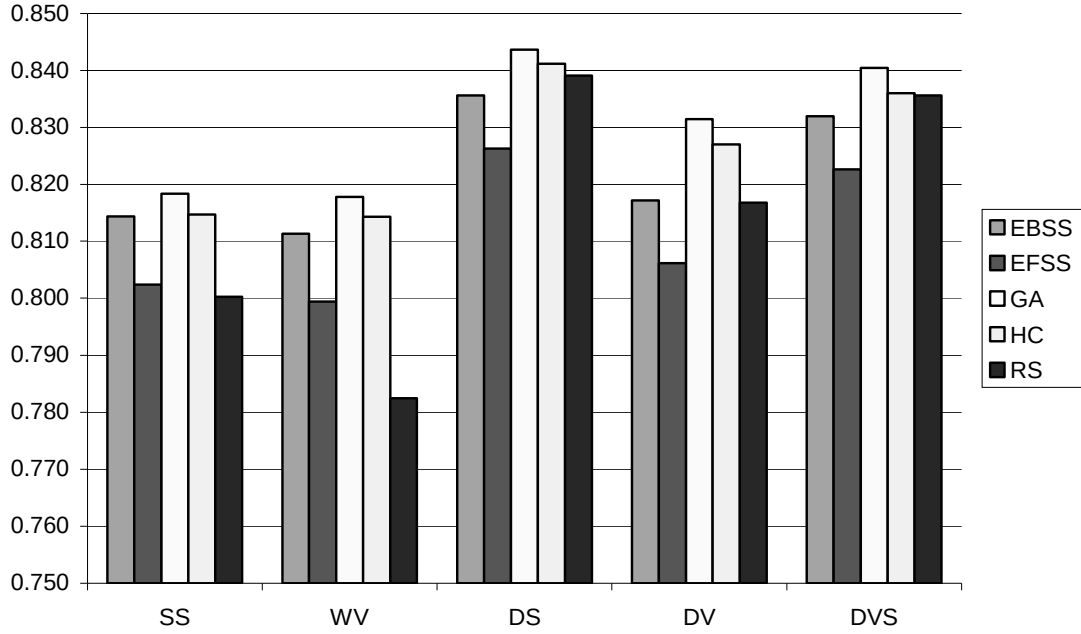


Figure 4. Average test set accuracy for different integration methods with different search strategies for data sets with the number of features less than 9 (10 data sets)

One can see from the figures the same behavior of the search strategies for the two groups, and the dependencies between them are almost the same. One interesting finding is that RS for the dynamic methods (DS, DV, and DVS) in the first group of data sets works relatively better than RS for the dynamic methods in the second group (in the first case only the GA works always better, while for the second group the GA, HC, and EBSS are always better than RS for the dynamic methods). This shows that RS works better with data sets having smaller numbers of features. However, surprisingly, for the static methods RS is relatively better for the second group, which is counterintuitive and may be due to the relatively lower diversity of ensembles, generated with RS for the data sets with bigger numbers of features. For the data sets with bigger numbers of features, the search strategies are able to achieve better diversities than RS, which is also supported by the difference between the accuracy of dynamic and static methods: it is about 2% for the first group and 3.5% for the second one.

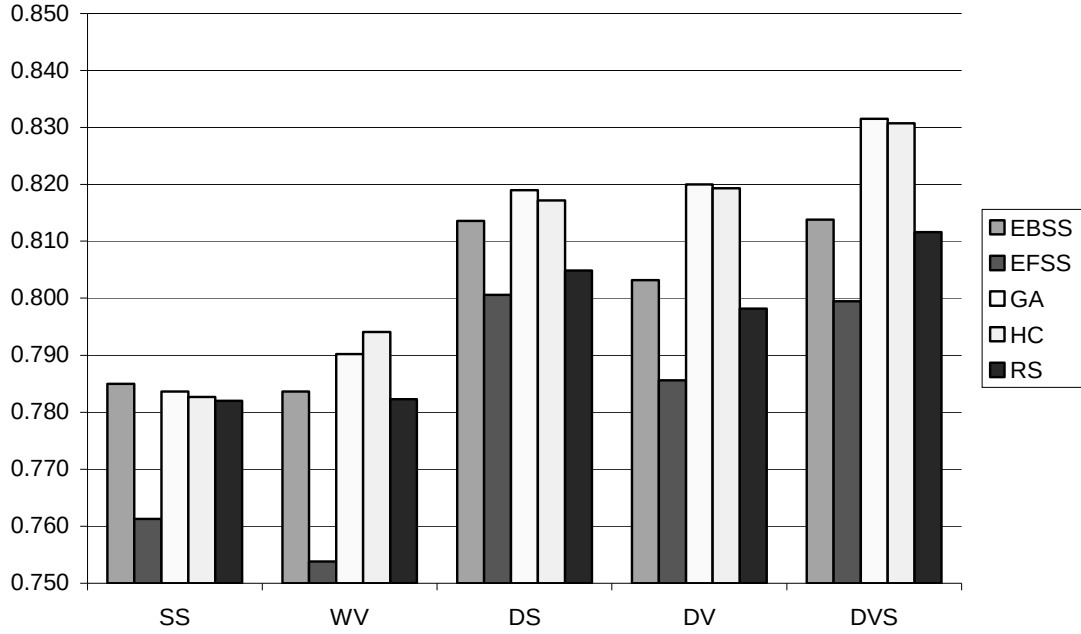


Figure 5. Average test set accuracy for different integration methods with different search strategies for data sets with the number of features more than or equal to 9 (11 data sets)

5.5 Integration methods and the GA search strategy

In Figure 6, the average ensemble accuracies are presented for the five integration methods for the GA strategy after 1, 5, and 10 generations. One can see from this figure that the static integration methods, SS and WV, and the dynamic DS start to overfit the validation data set after 5 generations and show lower accuracies, while the accuracies of DV and DVS continue to grow. This shows the importance of selection of the appropriate integration method for the GA search strategy.

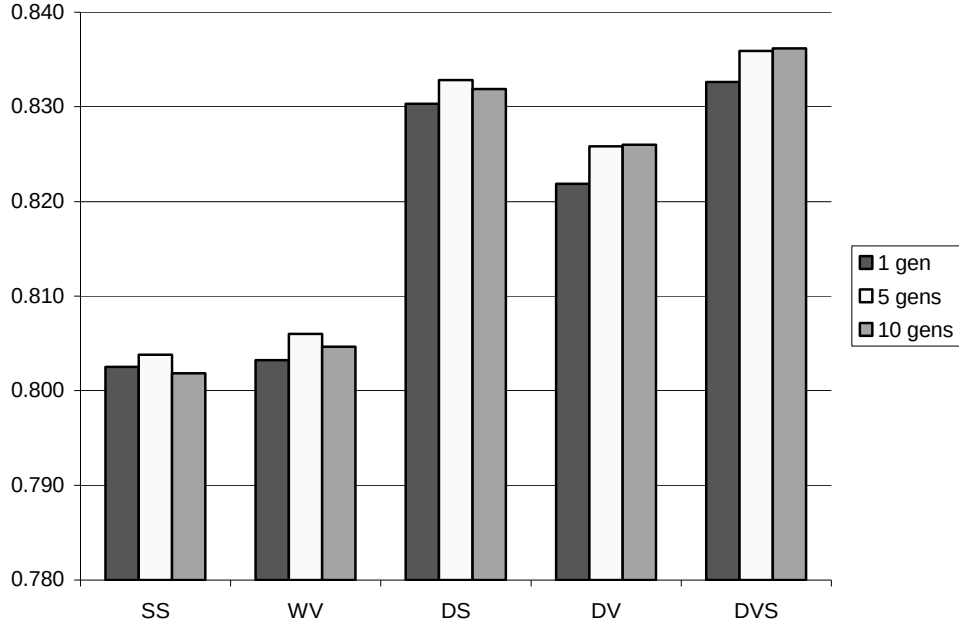


Figure 6. Average test set accuracy for different integration methods with the GA search strategy for different numbers of generations

5.6 Overfitting in the search strategies and integration methods

In Figure 7, as a measure of overfitting, we show the average ensemble accuracy on the test and validation sets correspondingly for the four search strategies with the DVS integration method. The RS ensemble accuracy is shown for the sake of comparison. As one can see from the figure, the highest difference between the test set and the validation set accuracies is with the GA and EFSS search strategies (0.038), the lowest difference in the test and validation set accuracies is with the HC search strategy (0.033), and with the EBSS search strategy the difference is equal to 0.036.

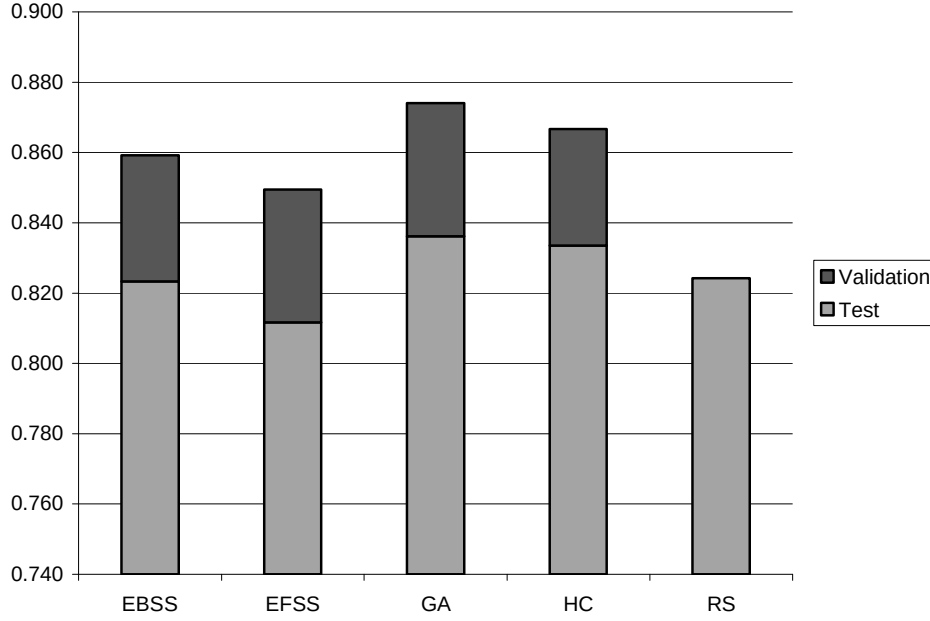


Figure 7. Overfitting for different search strategies with the DVS integration method (test set and validation set accuracies)

In Figure 8, as a measure of overfitting, we show the average ensemble accuracy on the test and validation sets correspondingly for the five integration methods, averaged over the search strategies. As one can see from the figure, the highest difference between the test set and validation set accuracies is with the WV integration method (0.038). Then we have DV (0.032), SS (0.031), DVS (0.029), and DS (0.027). In general, the static integration methods show more overfitting than their dynamic counterparts, as was also the case for the GA search strategy (Figure 6). It is worth noting that DVS returns the best accuracy on the unseen test data.

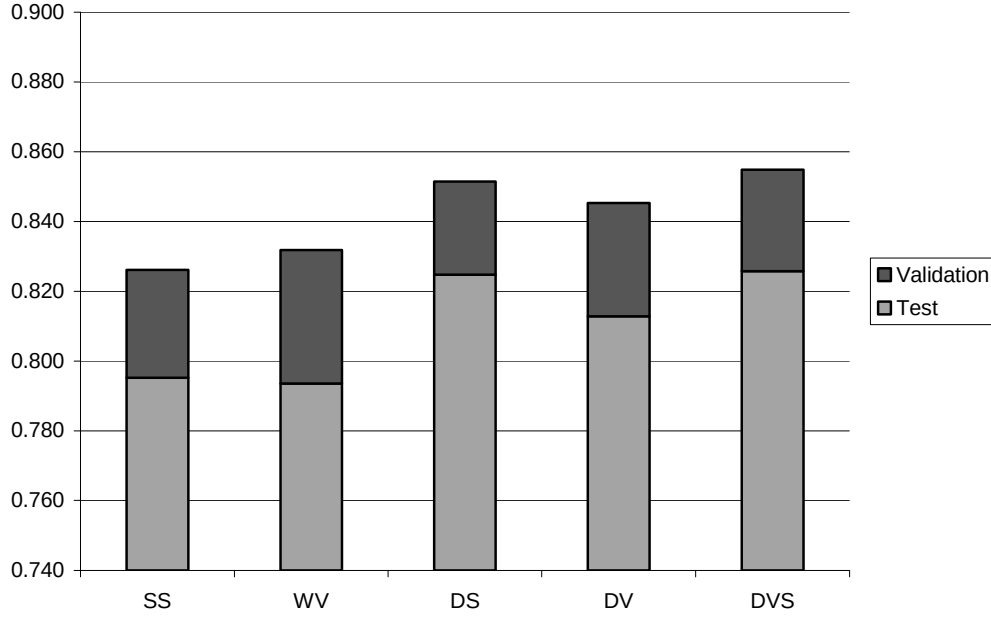


Figure 8. Overfitting for different integration methods averaged over the search strategies (test set and validation set accuracies)

5.7 Pre-selected *alpha* for integration methods and search strategies

In Figure 9, the pre-selected values of *alpha* are shown for the five integration methods and the four search strategies. Naturally, EBSS has the bigger values of *alpha* selected, as it needs more diversity to achieve better ensemble accuracy, when all the classifiers include mostly identical features. It is not the case with the static integration methods, especially WV, as diversity is not that important for them, as they do not use diversity to the same extent the dynamic integration does. In general, *alpha* for the dynamic methods is much bigger for all the strategies than for the static ones. The EFSS, HC, and GA strategies have almost the same *alpha* selected on average.

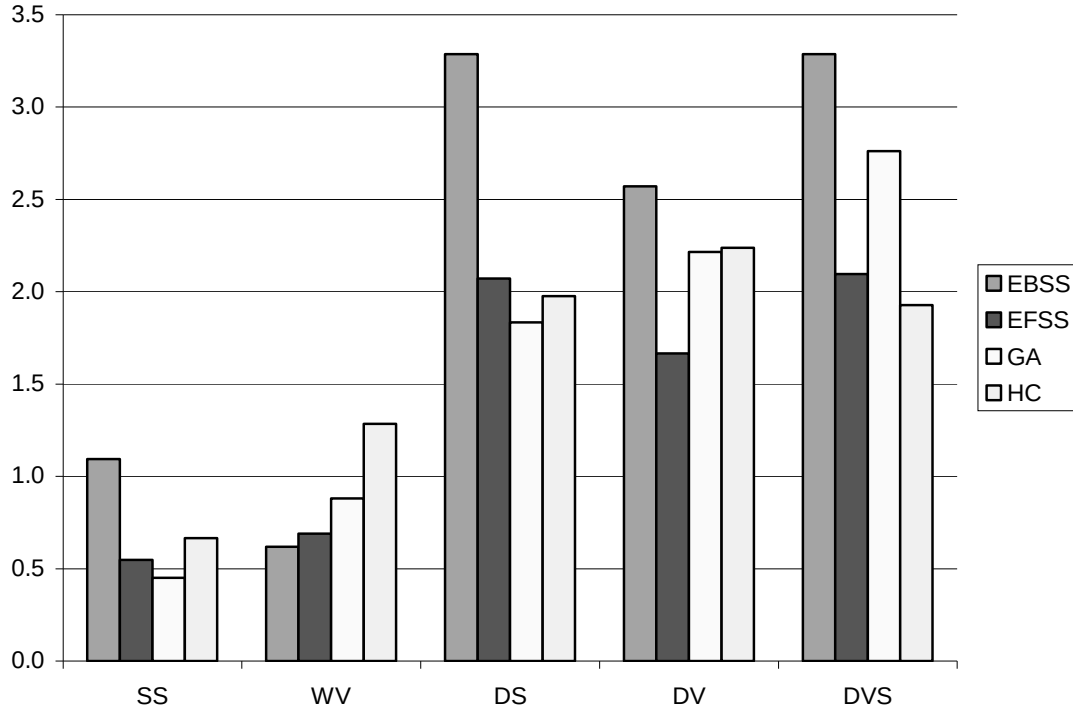


Figure 9. Average pre-selected alpha values for different integration methods and search strategies

5.8 Average numbers of features selected for different search strategies and integration methods

In Figure 10, the average relative numbers of features selected in the base classifiers for the four search strategies and RS and for the five integration methods are shown. These numbers of features correspond to the different *alpha* values selected for each combination of a search strategy, integration method, and data set (in the case of RS these numbers correspond simply to the random sets, and do not depend on the integration method). As one can see from the figure, naturally, RS selects exactly half of the features on average. Surprisingly, the GA strategy selects the biggest number of features, even higher than the EBSS strategy. One possible explanation for this is that, due to the global nature of its search, it is able to achieve better diversity even with classifiers including bigger feature subsets having more features in common. As expected, the EFSS strategy selects the lowest number of features on average. As a rule, more features are selected in the static integration methods than in the dynamic ones, corresponding to the bigger *alpha* values of Figure 9.

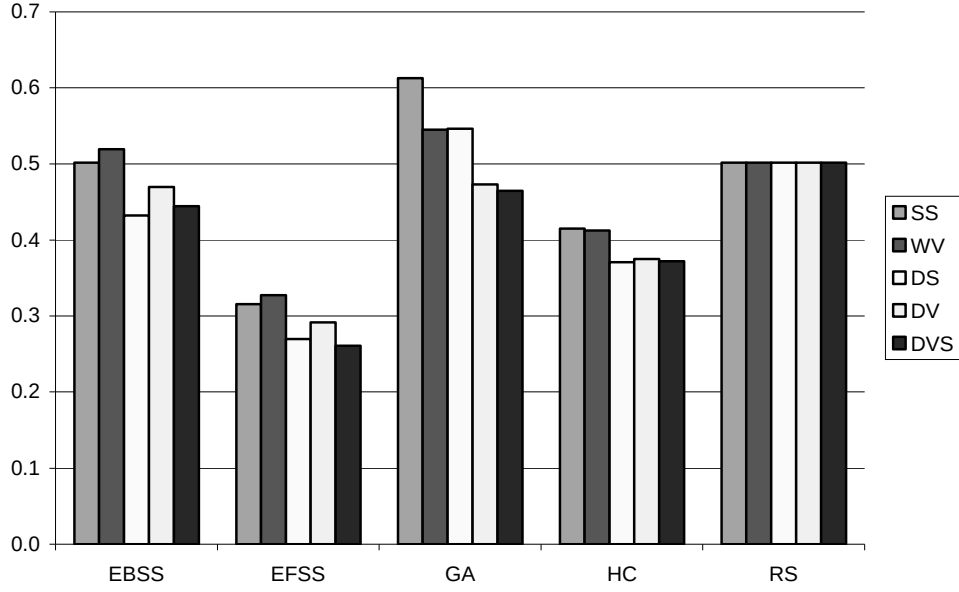


Figure 10. Average relative numbers of features in the base classifiers for different search strategies and integration methods

5.9 Pre-selected neighborhood for dynamic integration

In Figure 11, the average numbers of nearest neighbors (k) pre-selected for the dynamic integration methods are shown. As one can see from this figure, DS needs bigger values of k . This can be explained by the fact that its prediction is based on only one classifier being selected, and thus, it is very unstable. Bigger values of k provide more stability to DS. k is equal to 32 for DS on average, and it is only 10 for DV. For DVS, as a hybrid strategy, it is in between at 22. The selected values of k do not change significantly with the search strategies. The only “outlier” is the k value for DVS with the GA strategy. Such a high value can probably again be explained by the differences in the nature of the search between the GA and the other strategies.

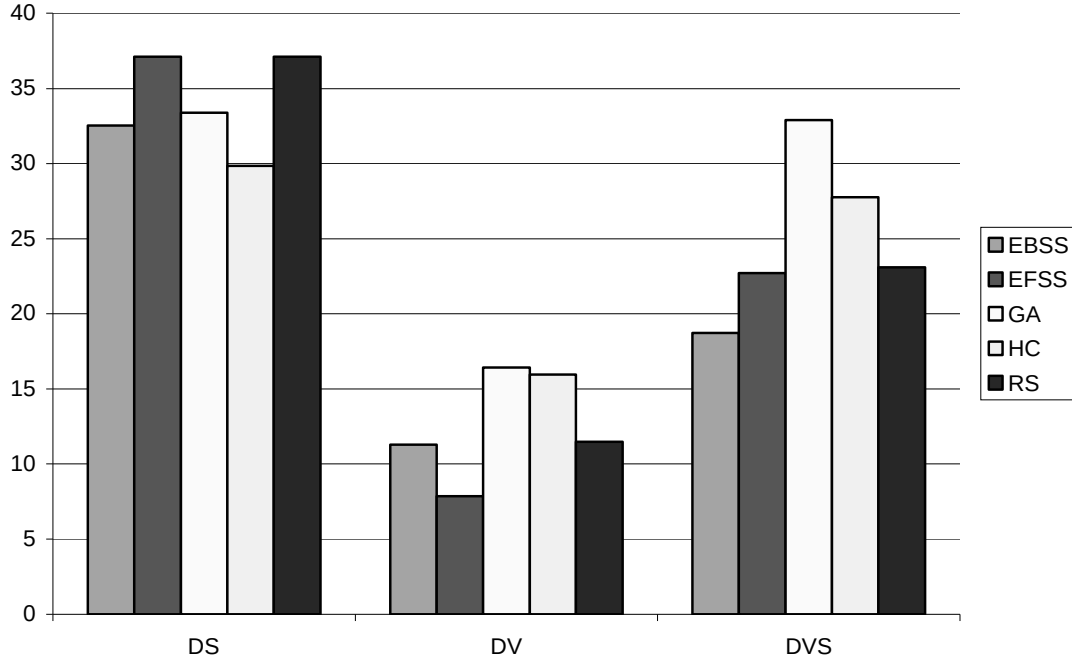


Figure 11. Average numbers of nearest neighbors pre-selected for the dynamic integration methods

Conclusions

In our paper, we have considered seven diversity metrics, five of which are pairwise and can be used as a component of the fitness function to guide the process of ensemble training in ensemble feature selection (the plain disagreement, *div_plain*; the fail/non-fail disagreement, *div_dis*; the *Q* statistic, *div_Q*; the correlation coefficient, *div_corr*; and the *kappa* statistic, *div_kappa*), and two non-pairwise (entropy, *div_ent*; and ambiguity, *div_amb*). All the seven measures can be used to measure the total ensemble diversity as a characteristic of ensemble goodness.

We consider four search strategies for ensemble feature selection: Hill Climbing (HC), a Genetic Algorithm for ensemble feature selection (GA), Ensemble Forward Sequential Selection (EFSS), and Ensemble Backward Sequential Selection (EBSS). In our implementation, all these search strategies employ the same fitness function, which is proportional to accuracy and diversity of the corresponding base classifier. Diversity in this context can be one of the five pairwise diversities considered. To integrate the base classifiers generated with the search strategies, we use five integration methods: Static Selection (SS), Weighted Voting (WV), Dynamic Selection (DS), Dynamic Voting (DV), and Dynamic Voting with Selection (DVS).

In our experiments, first, we analyse the total ensemble diversity calculated with the seven considered metrics. To check the goodness of each measure of diversity we calculated its correlation with the

difference between the ensemble accuracy and the average base classifier accuracy. The correlations did not depend on the search strategy used, and depend significantly on the data set. The best correlations averaged over the 21 data sets were shown by *div_plain*, *div_dis*, *div_ent*, and *div_amb*. *Div_Q* and *div_corr* behaved in a similar way, supported by the similarity of their formulae. Surprisingly, *div_Q* had the worst average correlation. All the correlations besides *div_Q* and *div_corr* changed with the change of the integration method, showing the different use of diversity by the integration methods. The correlation coefficients for *Div_amb* were almost the same as for *div_plain*.

Then, we have compared ensemble accuracies of the four search strategies for the five pairwise diversity metrics, and for the five integration methods being used. For all the diversities, the GA was the best strategy on average, and HC was the second best. The power of these strategies can be explained by the fact that they are based on random subsampling. Surprisingly, EFSS was significantly worse than the other strategies for this collection of data sets. There was no significant difference found in the use of the guiding diversities, besides the findings that *div_Q* was significantly worse on average than the other diversities for EBSS, EFSS, and the GA; and *div_kappa* was better than the other diversities for the GA. HC was not sensitive to the choice of the guiding diversity. The results of the Student's *t*-test for significance supported these findings. The dynamic integration methods DS, DV, and DVS always worked better on average than the static methods SS and WV, and DVS was the best dynamic method on average. The GA and DVS was the best combination of a search strategy and integration method.

To analyse overfitting in these ensembles, we have measured all the ensemble accuracies on both the validation and test sets. Among the search strategies, GA and EFSS had the biggest overfitting on average. For the integration methods, it was discovered that the static integration methods show more overfitting than their dynamic counterparts.

The experimental results presented change very much with the data set, and more data sets are needed to check and to analyse the reported findings. More experiments are needed to check how the behaviour of the diversity metrics depends on the data set characteristics, and what is the best metric for a particular data set. This is an interesting topic for further research.

Another measure of ensemble diversity, not considered in this paper, that has been shown to have high correlation with the ensemble accuracy is Double Fault (DF) measure [22, 34]. It is an interesting topic for future research to consider this measure also as the guiding diversity in our experimental setting, and to check its correlation with the improvement in accuracy, and with the considered measures of diversity.

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